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TOP 6-3-037 describes general procedures for technical testing of airborne target detection, acquisition, and tracking devices under conditions typical of those for which they were designed. Procedures include: inspection, installation, technical performance, logistics supportability, reliability, achieved availability, compatibility with other equipment, safety, human factors, and personnel and training requirements. Data requirements and presentation are also described.

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U.S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

AMSTE-RP 702-106
*Test Operations Procedure (TOP) 6-3-037 30 April 1990
AD NO. A223852

AIRBORNE TARGET DETECTION, ACQUISITION, AND TRACKING DEVICES

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1. SCOPE.

a. This TOP describes procedures for technical testing of airborne target detection, acquisition, and tracking devices under conditions typical of those for which they were designed.

b. Since the usage and design of these devices may cover such a wide range, the operational performance procedures have been presented in a general way to allow for satisfactory evaluation of devices of all types. The use of

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particular procedures will be determined by specific applications and locations. Pertinent, applicable requirements will be used to derive test criteria and to formulate test conditions.

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities. Testing should normally be conducted from fixed base airfields due to the sensitivity of equipment used to monitor, record, and analyze data.

2.1.1 Maintenance Area. Maintenance areas should be of sufficient size and configuration for parking, inspection, and limited repair of the test-bed aircraft. An area for unpacking, inspection, and installation of the test item and data collection equipment should be identified. Bench test facilities with power supplies should be available.

2.1.2 Ground Run/Test Flight Area. Ground run areas should normally be level, hard-surfaced, and close to instrumentation areas. Test flight areas should allow for operation under established flight profiles. Select test flight areas to exclude extreme heat, cold, or high altitude areas unless such conditions are a specific test consideration.

2.1.3 Maintenance Personnel, Tools, and Test Equipment. The complexity of the test-bed aircraft, test item(s), and data collection equipment will dictate the number and skill level of personnel selected to support the test. Select tools and test equipment under the same criteria. If military personnel of the appropriate MOS are not available, consider contractor's or manufacturer's support.

3. REQUIRED TEST CONDITIONS.

3.1 Support.

a. Select and schedule suitable operational areas at representative locations. Ensure availability of maintenance support facilities, special test equipment, repair parts, and personnel.

b. Ensure that all required support aircraft have been scheduled.

c. Obtain applicable frequency assignments, if required, before operating the test item and support equipment.

d. Coordinate with supporting and participating agencies, activities, and facilities.

3.2 Personnel.

a. Ensure that personnel with appropriate MOS and skill levels for operation and maintenance of the test item are properly trained.

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b. Provide additional training in the following areas:

(1) Procedures to be followed and data to be collected during the test.

(2) Hazards and safety precautions associated with the test item, test procedures, and test equipment.

c. If personnel are non-exempt troops IAW TECOM Supplement to AR 70- 25[1], ensure that the test plan has been reviewed and approved by the HQ TECOM Human Use Committee.

4. TEST PROCEDURES. The following subtests are arranged in a logical sequence to provide a step-by-step analysis of the suitability of target detection, acquisition, and tracking devices to perform their primary functions. In addition to the following subtests, consideration should be given to appropriate environmental tests from MIL-STD-810D[a], software testing, lightning strike, EMRO, EMRH, EMI/EMC (MIL-STD-461[b]/462[c]), nuclear and chemical effects survivability and electronic countermeasures and effects from friendly emissions.

4.1 Inspection. This subtest will establish condition and completeness of the test item and associated equipment.

4.1.1 Inventory and Visual Inspection.

a. Method.

(1) Inventory the test item against the Basic Issue Item List (BIIL).

(2) Inspect test item. Inspection points should include:

(a) Removal of all preservatives.

(b) Inspection for obvious defects.

(c) Completeness of assembly.

(d) Completeness of maintenance package and verification against any included packing list.

(e) Verification of any required lubrication.

(3) Inspect test item for safety hazards using a tailored safety and health checklist. Checklists in TOP 6-2-507[2] and 7-3-506[3], as appropriate, shall be used as a guide.

[1]Reference numbers/letters match those in appendix B.

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(4) Complete a Test Incident Report (TIR) for all observed shortages or discrepancies. (AMC Reg 70-13[4])

b. Data Required. Record the following:

(1) Physical defects of the test item and any shipping containers.

(2) Completeness of the BIIL.

(3) Completeness of assembly.

(4) Completeness of maintenance package.

4.1.2 Physical Characteristics. (TOP 7-3-500[5])

a. Method.

(1) Note the legibility, effectiveness, and correctness of markings, legends, etc.

(2) Determine the dimensions, weight, and volume (if applicable) of all test item components.

(3) Compute the test item's total weight and volume.

(4) Photograph all test item components before and after installation.

b. Data Required. Record the following:

(1) Equipment markings.

(2) Dimensions, volume, and weight from all components of the test item and the total system weight and volume.

4.1.3 Technical Characteristics.

a. Method. Examine the following technical properties (as applicable):

(1) Power requirements - Verify circuitry and measure the electrical power requirements of the test item. Check that the test item operates satisfactorily over the range specified in supply voltages by varying the supplies and noting performance.

(2) Controls, adjustments, and indicators (mechanical and electrical) - Determine the following as appropriate:

(a) Operation.

(b) Effect on the system.

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(c) Adequacy of movement.

(d) Proper calibration.

(e) Changes are monitored and displayed correctly.

(f) Range and sensitivity.

Evaluations requiring flight will be conducted during the Technical Performance Test.

(3) Equipment safety and protective devices - Determine adequacy.

(4) Fail-safe characteristics - Examine the system for the following:

(a) Internal failure - When the system becomes inoperative because of internal failure, operator personnel will be made aware of the condition. When possible, simulate failures and determine suitability of all fail-safe features.

(b) Acceptance or provision of external signals - If the system accepts/provides electrical signals from/to other on-board avionics equipment, operator personnel will be made aware of any out-of-limit level existing on any line. Simulate failures on each line and determine the adequacy of fail-safe indications.

(5) Confidence, self-checking or integrity circuits, if any - Verify proper operation.

(6) Primary technical characteristics - Measure the transmitter characteristics (frequency, power, repetition rate, pulse characteristics, etc.) and receiver characteristics (sensitivity and bandwidth).

(7) Cold start and warmup - Subject the system to the required number of cold start power application procedures. Determine warmup and effects due to multiple power-up applications.

b. Data Required. Record the following:

(1) Electrical incompatibility and power requirements.

(2) Controls and indicators.

(a) Improper operation.

(b) Calibration.

(c) Monitoring and display of system conditions.

(3) Operation of safety devices.

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(4) Operation of fail-safe features.

(5) Operation of self-checking features.

(6) Warmup time.

4.2 Installation. This subtest will determine installation requirements, including placement and removal procedures, stability of mountings, and flexibility for use on various aircraft.

a. Method.

(1) Determine installation requirements (time, effort, tools, materials, personnel, instructions, etc.).

(2) Ensure that the test item is secure, protected against shock and vibration, and mechanically stable.

(3) Determine mounting requirements for use on various types of aircraft.

(4) Photograph pertinent installation procedures.

b. Data Required. Record the following:

(1) Time required.

(2) Tools required.

(3) Ease of installation and removal.

(4) Equipment flexibility for arrangement and mounting.

(5) Accessibility for maintenance.

(6) Common (interchangeable) fittings and connections.

4.3 Technical Performance. This subtest will determine operation and performance characteristics of the test item.

4.3.1 Detection, Acquisition, and Tracking.

a. Method.

(1) Use a target of the type which the test item is designed to defeat.

(2) Fly toward, over, or by the target (depending on whether the test item is forward, downward, or side looking).

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(3) Determine the test item's ability to detect, acquire, and track the target.

(4) Note system performance (in particular, the display characteristics) during the period that the target is displayed and is being tracked.

b. Data Required. Record the following:

(1) Type of target used.

(2) Flight procedures used.

(3) Problems encountered in detecting, acquiring, or tracking targets.

4.3.2 Maximum System Range.

a. Method.

(1) Fly straight line flights at or over targets increasing the distance-to-target until the target is no longer distinguishable.

(2) Determine the maximum horizontal range (slant range) at which the test item is effective.

b. Data Required. Record the following:

(1) Test course and target information.

(2) Meteorological conditions.

(3) Aircraft altitude, heading, and speed.

(4) Maximum range for detection, recognition, and identification.

4.3.3 Target Position Accuracy.

a. Method.

(1) Determine the test item's ability to provide accurate coordinate information on targets. (Range to, angle with respect to aircraft axis, etc.)

(2) Conduct flights over a surveyed course which includes precisely located landmarks and targets.

(3) Fly various altitudes and angular ground tracks with respect to the targets so that on passage over landmarks, display information can be compared with actual target coordinates (corresponding to the flight conditions, e.g., altitude and angle taken with respect to the landmark).

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b. Data Required. Record the following:

(1) Test course and target information.

(2) Aircraft altitude, flight direction, display coordinate data at each checkpoint, and actual target coordinate data.

4.3.4 Target Resolution.

a. Method.

(1) Determine the test item's ability to resolve and adequately display targets in multiple configuration, i.e., targets of the same or varying size, shape, etc., with different separations. Include targets which will allow for elevation, azimuth, and range resolution, as appropriate.

(2) Fly, at constant radar altitude, approaches to or over the target groups and determine the range or altitude at which separation or resolution occurs.

(3) Use video tape, movie, or sequenced still photographs of displays during target approach to document the sequential resolving process.

b. Data Required. Record the following:

(1) Test course and target information.

(2) Aircraft altitude, heading, and speed.

(3) Distance at which separation or resolution occurs.

(4) Meteorological and target background.

4.3.5 Field of Coverage.

a. Method.

(1) Determine the extent and accuracy of the test item field of coverage, i.e., the system-displayed indicated position of a target compared to the actual position (at the maximum of each coordinate available on the display). Include the following where appropriate:

(a) Maximum elevation above and below aircraft axis.

(b) Azimuth left and right of aircraft axis.

(c) Angular or circular field of coverage in the plane below the aircraft.

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(2) Conduct straight and level flights at various angles and altitudes which will place targets at the extremity of each display coordinate. Accurate aircraft position information is required.

b. Data Required. Record the following:

(1) Test course and target information.

(2) Aircraft altitude, heading, and speed.

(3) Position data on the aircraft when targets are lost at the extremities of the display.

4.3.6 Target Characteristics/Environment.

a. Method.

(1) Operate the test item against various representative military targets to determine capabilities and limitations of the test item as related to variations in target characteristics and their environment and backgrounds.

(2) Include the following types of targets in the tests:

(a) Boats.

(b) Aircraft.

(c) Trains.

(d) Man-made structures.

(e) Troop concentrations and encampments.

(f) Military vehicles.

(3) Vary the target characteristics and backgrounds to determine effects on test item performance and displays. Include the following:

(a) Moving and fixed.

(b) Target temperatures.

(c) Background terrains.

(d) Camouflaged and uncamouflaged.

(e) Targets varying in number, geometry, and spacing.

(f) Targets constructed of different materials.

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(4) Repeat selected tests during periods when changes in atmospheric conditions exist. Include the following conditions:

(a) Sunrise and sunset.

(b) Daytime and nighttime hours.

(c) Rain, fog, snow, and clouds.

(d) High and low temperatures.

(5) Conduct tests, day and night, over various types of geographic surfaces. Consider the following:

(a) Water/land interfaces.

(b) Isolated bodies of water.

(c) Mountain ranges.

(d) Areas with varying types and degrees of vegetation.

(e) Snow-covered ground, desert, etc. (look at extremes).

b. Data Required. Record the following:

(1) Flight procedures used.

(2) Target characteristics.

(3) Target background.

(4) Variations in displays.

(5) Atmospheric conditions at the time of data collection.

(6) Changes in area coverage, accuracy, range, etc.

(7) Changes in display of targets.

(8) Details of the geographic surface at the time of data collection.

(9) Comments regarding effectiveness of the test item and any problems experienced.

4.3.7 Displays.

a. Method. Document the following using photographic techniques or some other suitable method of recording:

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(1) Stability/distortion.

(2) Readability.

(3) Definition and resolution.

(4) Correlation of display with known target characteristics.

b. Data Required. Record the following:

(1) Comments on the general effectiveness of displays.

(2) Correlation of display with targets.

(3) Suggestions for improvements.

4.4 Logistic Supportability. Testing will normally be conducted simultaneously and in conjunction with other test operations. Separate, independent test functions, real or simulated, will be performed as required to ensure a complete exercise of all the logistic supportability aspects of the test system. The scope of the logistic supportability testing will encompass all subelements listed below in accordance with TOP 7-3-507[6]. Testing must be consistent with the availability of required support elements and the maturity of the test hardware. These data should be delineated in the test planning/execution directive.

- End item requirement
- Supply support
- Technical data/equipment publications
- Support and test equipment
- Manpower and personnel, training, and training devices
- Transportation and handling
- Facilities

These basic subelements may be further subdivided to enhance the clarity and understanding of an individual subtest. Subelement breakouts/divisions are usually dependent upon the maturity and complexity of the test system and test constraints (time, dollars) placed on the test effort. Specific criteria for each subelement test must be extracted from program documentation (requirements, specification, acquisition, etc.). A brief explanation of each logistic supportability test subelement listed above and specific guidance to conduct each logistic supportability subelement test are contained in TOP 7-3-507.

4.5 Reliability. This subtest will determine reliability of the test item in terms of failure-free operation time and mean-time-between-failure (MTBF).

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a. Method.

(1) Maintain a complete log of all assembly, installation, operation, disassembly, and maintenance performed on components and the complete test item.

(2) Record data on Supportability Analysis Charts (SACs) as described in TOP 7-3-507.

(3) Calculate the MTBF of the test item, using appropriate procedures in TOP 7-3-508[7].

b. Data Required. Record the following:

(1) Hours of operation, daily and cumulative.

(2) Equipment failures and malfunctions, including chronological data required to determine failure-free operating time and MTBF.

(3) Completed SACs.

(4) Effect of failures on test conduct.

4.6 Achieved Availability. This subtest will determine the probability that the test item, when used under stated conditions in an ideal support environment, will operate satisfactorily at any given time (TOP 7-3-507).

a. Method.

(1) Calculate the Operating Time (OT) of the test item.

(2) Calculate the Preventive Maintenance Time (PMT) devoted to the test item.

(3) Calculate the Corrective Maintenance Time (CMT) devoted to the test item.

(4) Calculate the Achieved Availability (Aa) of the test item using the following formula:

$$Aa = \frac{OT}{OT + PMT + CMT}$$

b. Data Required. Use the data previously collected on maintainability, reliability, and operating time.

4.7 Compatibility. This subtest will determine effects and interactions of the test item with the established configuration of the test-bed aircraft. (TOP 7-3-509[8])

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a. Method. Note any instance of incompatibility using the following as a guide:

(1) Preparation for use - Note the need, during installation, for special tools, hardware, mounting brackets, etc., and nonstandard size or overweight items.

(2) Operation and performance - Note, during operational procedures, whether the test item interferes with, or is interfered by, other on-board or ground-based electronic systems (EMI/EMC testing). This can be accomplished by operating the test item simultaneously with other on-board electrical systems and in the presence of various types of radio frequency fields generated by ground installations. Compatibility of the test item with other on-board electronic systems using signals from or providing signals to the test item will be determined.

b. Data Required. Record the following:

(1) Physical or electrical characteristics not compatible with system capabilities.

(2) Instances when the test item interferes with or is interfered by other equipment generating, using, or radiating electrical energy.

(3) Test item compatibility with existing electronic equipment.

4.8 Safety. This subtest will determine hazardous characteristics of the design and operating procedures of the test item (TECOM Supplement 1 to AR 385-16[9], MTP 7-3-506).

a. Method.

(1) Observe proper safety precautions and adhere closely to any manual's directives which deal with safety and/or protection.

(2) Examine the procedures for all tests. Report any hazard to project personnel.

(3) Examine the characteristics of the test item including procedures for its operation and maintenance to ensure that maximum safety has been provided. Consider the following:

(a) Review airworthiness/safety releases required for flight.

(b) Examine operating procedures with a view that improperly executed or misinterpreted instructions could result in bodily harm or equipment damage.

(c) When hazardous conditions cannot be avoided, is the item properly and conspicuously marked for the conditions?

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(d) Are all moving parts shielded and completely enclosed?

(e) When electrical power is used, are the electrical circuits guarded against accidental contact?

(f) Are environmental limitations explicitly denoted?

(4) Determine the adequacy of all protective and warning devices. Consider the following:

(a) Overheat devices.

(b) Overload protection.

(c) Locking mechanisms.

(d) Limit switches.

(e) Visual and audible warning devices.

(f) Interlocks.

b. Data Required. Record the following:

(1) Comments regarding hazardous conditions found during the conduct of any test.

(2) For general safety characteristics:

(a) Poorly worded or unclear operating instructions.

(b) Warning labels lacking or not conspicuous.

(c) Unprotected electrical circuits.

(d) Markings of environmental limitation missing.

(3) Prepare a table to include the following:

(a) A list of all safety devices used.

(b) The type of failure each device is to detect.

(c) Indication that the device has successfully operated.

(4) List any missing devices or unsafe conditions.

(5) List any suggested additions to the safety features.

(6) Prepare a safety confirmation statement.

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4.9 Human Factors. This subtest will evaluate the degree to which the test item conforms to acceptable human engineering design criteria outlined in MIL-STD-1472D[10] and MIL-HDBK-759A[11].

a. Method. The human factors characteristics of the system will be assessed using task and design characteristics checklists and developed in accordance with TECOM TOP 1-2-610[12]. The purpose of the checklist is to rate, from a human factors standpoint, the tasks associated with, and the characteristics revealed during the procedures for preparing, operating, transporting, and maintaining the test item. The rating will be either satisfactory or unsatisfactory with explanatory information accompanying an unsatisfactory rating. Subjective data from personnel performing operation/maintenance tasks outlined in the instruction/technical manuals will be collected using questionnaires developed and administered in accordance with TECOM Pam 602-1[13]. Questionnaire data would be needed in paragraphs 4.3.6.b(9) and 4.3.7.b(1).

(1) For all tasks/characteristics the following will be considered:

(a) Ease of performance - Mental and physical effort required.

(b) Support -Adequacy of instructions and tools for the task.

(c) Time required - Modification of procedures to reduce time required.

(d) Design characteristics - Effects on performance of tasks.

(2) Rate the following tasks/characteristics for the evaluation listed.

(a) Preparation for use.

Assemble components, move to installation location, place in position, make external connections, and lock into position with fasteners, connectors, etc.

Apply power, check controls and indicators.

Make required alignment, calibrate, and adjust.

(b) Operation and performance.

Operate controls, note changes in equipment status, monitor other displays.

Evaluate interface between test item controls and indicators and aircraft controls and indicators.

Note legends, their effectiveness, readability, visibility.

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Note performance, operation, and system status feedback to operator (auditory, visual, etc.).

Night operation - adequacy of the test item's control and indicator lights and interface with cockpit lighting scheme.

(c) Maintenance.

Perform preventive maintenance:

Clean, add lubricants.

Remove and replace minor items.

Tighten fasteners, connectors.

Adjust, calibrate, align.

Perform unscheduled maintenance:

Detect malfunction by observing displays, noting visual or audible changes, or changes in operating effectiveness.

Isolate and identify malfunction causes by visual means or instrumentation.

Open, gain access to, and remove component.

Replace or repair and reestablish proper operation.

b. Data Required. Record instances of the following:

(1) Instructions.

(a) Lacking clarity.

(b) Insufficient or excessive detail.

(2) Tools.

(a) Proper tools not supplied.

(b) Excess of special tools specified.

(c) Additional tools recommended.

(3) Mental and physical effort.

(a) Above average skill or strength required.

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(b) Task is excessively tiring.

(4) Design.

(a) Poor location of component.

(b) Component not accessible.

(c) Visibility hindered.

(d) Adequacy of controls, adjustments, etc.

(5) Time required for task (if excessive, reason why).

(6) Personnel requirements.

(a) Insufficient number specified.

(b) Qualifications in error.

4.10 Personnel and Training Requirements. This subtest will determine adequacy and sufficiency of training and appropriateness of skill levels and experience required to operate and maintain the test item.

a. Method.

(1) Employ personnel of varying skill levels and experience throughout the test program to determine skill level requirements.

(2) Review performance of personnel during operational and maintenance procedures paying particular attention to mistakes or errors made in operational procedures and excessively long maintenance tasks.

(3) Review the effects of training programs as to their adequacy, etc., noting also any additional training required during the test and suggestions for changes to the training program.

(4) Make a quantitative estimate as to the average number of training hours required for both operational and maintenance personnel.

b. Data Required. Record the following:

(1) The most appropriate skill levels and experience (background) suggested for operator and maintenance personnel.

(2) Any suggested changes of training techniques, literature, etc., to eliminate operator errors or reduce maintenance time.

(3) Each additional training technique used after the start of testing and suggestions for additions or deletions to the training program.

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(4) The training time required for maintenance and operator personnel.

5. DATA REQUIRED.

5.1 Preparation for Test. Data to be recorded before testing will include, as a minimum, the following:

a. Test item nomenclature, serial number(s), and other pertinent data.

b. Test equipment nomenclature, serial number(s), functional description, accuracy tolerances, calibration requirements, and date of last calibration.

5.2 Test Conduct.

a. Refer to paragraph 4, Test Procedures, for specific data requirements for individual subtests.

b. Data originating in all tests and subtests will be recorded as appropriate:

(1) Operators', observers', and test controllers' records and questionnaires.

(2) Narrative comments and observations.

(3) Maintenance records.

(4) Photographs.

(5) Diagrams and sketches.

c. All data will be properly identified and annotated with respect to:

(1) Test/subtest.

(2) Source.

(3) Date/time.

(4) Pertinent information.

d. Data to be recorded in addition to specific instructions given in preceding paragraphs will include:

(1) Sample size (number of measurement repetitions).

(2) Instrument or measurement system mean accuracy tolerances.

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6. PRESENTATION OF DATA. Test results will consist of the following:

a. Composite documentation of the reduced and correlated data arranged by subtest in the general form of narrative description supported by diagrams, photographs, graphs, and tabular data.

b. An analysis to determine the extent to which the test item and its maintenance package meet stated requirements and detailed military specifications.

c. Supplements or annexes to the basic report, delineating the common test factors which are of sufficient scope, importance, and/or complexity to warrant separate treatment. Each supplement will include the applicable supporting data.

d. A further analysis to determine the extent to which the test item exceeds the performance characteristics or otherwise provides distinct advantages over existing Army equipment providing the same functions.

e. Equipment evaluation will be limited to comparing the actual test results to the equipment specifications imposed by the intended usage.

Forward comments, recommended changes, or any pertinent data which may be of use in improving this publication to Commander, U.S. Army Test and Evaluation Command, ATTN: AMSTE-TC-M, Aberdeen Proving Ground, MD 21005-5055. Technical information may be obtained from the preparing activity: Commander, U.S. Army Aviation Development Test Activity, ATTN: STEBG-MP-P, Fort Rucker, AL 36362-5276. Additional copies are available from the Defense Technical Information Center, Cameron Station, Alexandria, VA 22304- 6145. This document is identified by the accession number (AD No.) printed on the first page.

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APPENDIX A. BACKGROUND

1. Army aircraft used in missions such as surveillance and support for ground operations will be enhanced from a tactical military standpoint when they are equipped with target detection, acquisition, and tracking devices. These systems allow flight personnel to determine the presence and positions of targets, analyze their characteristics, track motion, and direct weapons fire. These functions would be impossible when visual sightings become restricted. Targets can be landmarks, vehicles, personnel, encampments, storage areas, etc.

2. The basic components of a system of this type are:

a. Illuminator - (transmitter) - generates the signal to be transmitted.

b. Antenna - shapes and directs the transmitter energy and collects reflected or target emitted energy.

c. Receiver - processes information contained in received energy.

d. Display - provides target information in some desired form (position, size, etc.).

3. A major subdivision occurs in this class of systems when there is - or is not - a transmitter. Systems with transmitters are classed as active; those without are passive. In operation, the system receiver collects either reflected transmitter energy or energy emitted directly by the target, e.g., infrared. Typical examples of system types include:

a. Low light level television (LLLTV) - passive.

b. Laser systems - active.

c. Microwave radar - active.

d. Infrared (side, forward, and downward looking) - passive.

4. Within the class of target detection, acquisition, and tracking systems, there is a wide range in the technical and operating characteristics of a given system. In general, an individual system is designed, and will have been selected for, applications involving targets and conditions for which it is expected to be most effective. However, for the purpose of determining what test procedures and conditions should be utilized in formulating a test plan, it is desirable to consider the subject as divided into three aspects: the system and its characteristics; the medium between the system and targets and its effects on propagation; and the target/target background combination characteristics and effects. These aspects will be treated in detail during operational tests. The primary intent, however, for the technical test of these systems will be to show that the system provides reliable and accurate target information while operating through various mediums against targets

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with varying characteristics against different backgrounds. Testing of the system also must ensure reliability and safety since the system will be exposed to many different environments while the aircraft is involved in flight procedures. Other factors such as flexible installation and physical characteristics rendering the system suitable for use on different aircraft also will be examined. The total testing program shall verify the suitability of the device for use by the Army.

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APPENDIX B. REFERENCES

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